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23 February 1982
SUPERSEDING
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10 May 1974

MILITARY SPECIFICATION

# FANS, CENTRIFUGAL, INDUSTRIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers electric-motor-driven centrifugal fans which are not integral parts of equipment designed as complete units by manufacturers, and which are designed for ventilating uses and industrial air-handling applications.

* 1.2 Classification. Fans shall be of the following types, as specified (see 6.2.1):

Type I - Forward-curved blades.
Type II - Radial blades.
Type III - Backward-inclined blades.

## 2. APPLICABLE DOCUMENTS

2.1 *Government documents.*

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).
ZZ-B-225 - Belt, V, Industrial (Multiple Drive).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4140

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MILITARY

MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (for the Treatment of Communications, Electronic and Associated Equipment).
MIL-F-3541 - Fitting, Lubrication.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts; Packaging of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics; Measurement of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF LABOR (DOL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards with Supplemental Revisions.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20406.)

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA)

AMCA Publication No. 99.
AMCA Standard 210 - Laboratory Methods of Testing Fans for Rating Purposes.
AMCA Standard 300 - Test Code for Sound Rating.

(Application for copies should be addressed to the Air Movement and Control Association, 30 West University Drive, Arlington Heights, IL 60004.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B1.1 - Unified Screw Threads.
B2.1 - Pipe Threads (Except Dryseal).
B2.2 - Dryseal Pipe Threads.
B15.1 - Mechanical Power - Transmission Apparatus, Safety Code for.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)



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NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

ICS - 1 - General Standards for Industrial Controls and Systems.
ICS - 2 - Industrial Control Devices, Controllers and Assemblies.
ICS - 6 - Enclosures for Industrial Controls and Systems.
MG - 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 2101 L Street, N.W., Washington, DC 20037.)

RUBBER MANUFACTURERS ASSOCIATION (RMA)

Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to the Rubber Manufacturers Association, 1901 Pennsylvania Avenue, N.W., Washington, DC 20006.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

## 3. REQUIREMENTS

3.1 Description. The unit shall consist of a centrifugal fan, electric motor, controls, guards, and accessories. All components shall be attached to a common base which shall include provisions for fastening it to a foundation. The unit shall be completely assembled, ready for installation.

* 3.2 First article. When specified (see 6.2.1), the contractor shall furnish a fan of the type required for first article inspection and approval (see 4.2.1 and 6.3).

* 3.3 Standard commercial product. Each fan of the same classification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the fan being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

* 3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.



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* 3.5 Identical items. All fans of the same classification, furnished under any specific contract, shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviation will be acceptable without prior written approval of the contracting officer.

* 3.6 Standards compliance. Fans shall be capacity rated in accordance with AMCA Standard 210 and sound rated in accordance with AMCA Standard 300, prior to approval of the first shipment, if one is specified, or prior to approval of the first shipment, if a first article is not specified. The supplier shall submit to the contracting officer satisfactory evidence that the fan he proposes to furnish has been rated in accordance with AMCA 210 and 300. Acceptable evidence of meeting the requirements of these standards will be the AMCA Certified Rating Seal, or certified test reports (see 6.2.2) from an independent testing laboratory indicating that the fan ratings conform to the requirements of AMCA 210 or to the capacity test of 4.6.2, and AMCA 300.

* 3.7 Design. The fan assembly shall be designed to handle clean or dust-laden air at temperatures up to 250°F, and up to 600°F when equipped with a bearing cooling device (see 3.11.3.1). The design of the equipment and accessories shall permit accessibility for maintenance and service in the field. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment. The fan assembly shall be built in accordance with AMCA Publication No. 99.

* 3.8 Safety. All parts which are energized electrically shall be insulated, fully enclosed, or guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All moving parts hazardous to personnel shall be enclosed or guarded. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. Protective devices shall not impair the operating functions. Warnings shall be mounted on or near components containing hidden hazards. Applicable portions of OSHA 1910.219 shall apply.

* 3.9 Maintainability. The fans shall operate as specified herein, without maintenance other than the supplier's recommended normal scheduled adjustments and servicing as established by a maintenance schedule prepared and submitted by the supplier prior to test. All major assemblies shall be accessible for maintenance, repair, and replacement. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be located for accessibility. All fasteners shall be of corrosion-resistant material or shall be treated to be corrosion resistant. All screw-threads shall be in accordance with ANSI B1.1, B2.1 and B2.2, as applicable. Maximum use shall be made of interchangeable hardware and fastening devices.

* 3.10 Performance. The fan shall deliver its rated capacity in cubic feet per minute at the static pressure in inches of water specified (see 6.2.1). Fan sound power level shall not exceed 90 decibels.

3.11 Details of components. The unit shall consist of the components specified below; and when the motor is connected to the matching power source, the unit shall function in accordance with the performance requirements.



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* 3.11.1 Fan wheel. The fan wheel shall be constructed of cast aluminum alloy or carbon steel plate, as specified (see 6.2.1). Each steel blade shall be riveted (see 3.20.3) or welded (see 3.20.4) to form the fan wheel. The complete rotating assembly shall be dynamically balanced within the limits of the following formula:

$$\text{Vibration displacement, Mils - peak-to-peak} = \frac{1{,}620}{\text{revolutions per minute}}$$

* 3.11.2 Fan shaft. The shaft shall be turned, ground, and polished carbon steel alloy. The maximum allowable runout after all machining shall be 0.001 inch on shafts up to 1.5 inches in diameter and 0.002 inch on shafts greater than 1.5 inches in diameter. The shaft shall be sized for maximum speed of the size of fan involved and shall operate at not more than 80 percent of the first critical speed.

* 3.11.3 Fan bearings. Unless otherwise specified (see 6.2.1), the fan shall be equipped with precision anti-friction bearings conforming to FF-B-171 (with an average life in accordance with manufacturer's standard). The bearings shall be of the self-alining type and shall be mounted in suitable pillow blocks having provision for attachment to the fan housing or base assembly.

* 3.11.3.1 Bearing cooling device. When specified (see 6.2.1), a bearing cooling fan of suitable size, of split design, made of cast aluminum or a heat slinger shall be provided with anti-friction bearings when the air temperature in the fan is greater than 250°F but not less than 600°F.

* 3.11.4 Fan housing. The housing shall be constructed of carbon steel plate or sheet and of a nominal thickness not less than that listed in Table I. The scroll and sideplate joints shall be made airtight. The housing shall be reinforced to minimize vibration. When specified (see 6.2.1), the discharge direction for types I and III larger than 33-inches fan diameter and type II larger than 57-inch fan diameter, shall be easily converted at angles of 45 degrees. When specified (see 6.2.1), a threaded drain connection to accommodate a one-inch or 3/4 inch standard pipe, as specified (see 6.2.1), standard pipe shall be located at the lowest point in the scroll. When specified (see 6.2.1), a flush type access door shall be installed on the fan scroll and held by quick-release clamps on type II fans. Inlet and outlet duct connections shall be flanged.

TABLE I. Housing gages.

| Nominal fan diameter (inches) | Sides and Scroll |
|---|---|
| 12 up to 16 | 16 gage |
| 16 up to 24 | 14 gage |
| 24 up to 72 | 12 gage |
| 72 up to 132 | 10 gage |

* 3.11.5 Motor. The motor shall conform to NEMA MG-1, and shall be capable of continuous operation when driving the fan at rated conditions, and may be single or multi-speed, unless one of these is definitely specified (see 6.2.1). Features of bearings, enclosure, and electrical characteristics shall



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be as specified (see 6.2.1). When specified (see 6.2.1), an across-line partwind wye-delta or autotransformer shall be supplied. When specified, the motor shall be open drip proof, totally enclosed, or explosion proof (see 6.2.1).

* 3.11.5.1 Direct drive. When a direct drive is specified (see 6.2.1), the fan wheel shall be mounted on the motor shaft or driven through a suitable coupling by the motor. The motor shall be alined to minimize drive line vibration. When specified (see 6.2.1), the assembly shall be mounted on resilient vibration isolators.

* 3.11.5.2 V-belt drive. The fan shall be driven by means of single or multiple V-belts. Quantity and cross-sectional size of the belts shall be determined by the motor horsepower rating corrected to the service and arc-of-contact factors. For the motor location specified, a standard belt length shall be used to the nearest, but not less than the recommended, motor and fan shaft center distance. Sheave pitch diameters shall be as specified by the manufacturer for the speed ratio and fan speed required. When specified (see 6.2.1), sheaves shall be of a variable pitch type. The motor shall be solidly mounted to a suitable base which shall provide for belt tension adjustment and for anchorage to the fan foundation. All exposed belt driven fans shall have safety guards that conform to ANSI B15.1 and adjustable motor mounts for belt tension adjustment and belt replacement. V-belts shall conform to ZZ-B-225. Sheaves shall conform to NEMA MG-1 for minimum electric motor sheave diameter for a given hp motor. Motor sheaves shall be of the adjustable pitch type which provides a 20 percent speed variation, selected so that the specified capacity will be delivered with sheaves at approximately mid-position. V-belt drives shall be selected for continuous duty in accordance with Engineering Standards of Rubber Manufacturer's Association with a service factor of 1.5.

* 3.11.5.3 Motor starter. The starter control shall be manual or magnetic as specified (see 6.2.1), applicable NEMA type with thermal overload protection of the size and capacity suitable for the required service (see NEMA ICS-1, ICS-2). The control enclosure shall be as specified (see NEMA ICS-6 and 6.2.1).

3.12 Guards.

* 3.12.1 Inlet guard. Unless otherwise specified (see 6.2.1), the type I and type III fans shall be provided with an air-inlet screen guard of heavy gage galvanized wire. The meshes of the screen guard shall not be greater than 2 inches square and not less than 1/2 inch square. If required to facilitate removal, the screen shall be of split construction.

* 3.12.2 V-belt guard. A protective guard of heavy gage wire or sheet metal shall be mounted to the fan unit to prevent objects from being jammed between the belts and sheaves.

* 3.12.3 Shaft and coupling guard. A protective guard of heavy gage wire or sheet metal shall be mounted on the fan unit to completely enclose the motor shaft, coupling, and fan shaft.

* 3.12.4 Bearing-cooling fan guard. When a bearing-cooling fan is supplied (see 3.11.3.1), a protective guard of heavy gage wire mesh shall be mounted on the fan unit to completely enclose the bearing-cooling fan.



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3.13 Electromagnetic interference suppression. When specified (see 6.2.1), the unit shall conform to the electromagnetic interference suppression requirements of MIL-STD-461, Class C3, Group 1.

3.14 Lifting attachments. When specified (see 6.2.1), suitable lifting-attachment plate(s) shall be provided to enable the equipment to be lifted in its normal position. Attachment eye(s) shall be not less than three inches in diameter, and plate(s) shall withstand any handling conditions encountered, such as rapid lowering and braking of the load. When practicable, only one attachment shall be used. Where more than one is required, each attachment, when practicable, shall be of sufficient capacity to carry the total weight. A copper or zinc-base-alloy or aluminum plate securely fastened to the unit shall indicate diagrammatically the lifting capacity of each attachment.

* 3.15 Lubrication. Unless otherwise specified (see 6.2.1), means for lubricating shall be provided in accordance with the manufacturer's standard practice. Parts requiring lubrication shall be so located as to make the lubricating points easily visible and accessible. All parts requiring lubrication shall be properly lubricated before delivery. Grease fittings shall conform to the requirements of MIL-F-3541.

* 3.16 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

* 3.17 Cleaning, treatment, and painting. Unless otherwise specified (see 6.2.1), surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 1.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

* 3.18 Technical manuals. The contractor shall provide technical manual(s) conforming to the Data Item Description(s) (DID) listed on the Contract Data Requirements List (CDRL) (DD Form 1423), (see 6.2.2).

3.19 Dissimilar metals. Metals dissimilar with respect to the galvanic scale shall not be used unless separated by an insulating material which will avoid electrolytic corrosion.



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* 3.20 Workmanship.

* 3.20.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

* 3.20.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

* 3.20.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

* 3.20.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

* 3.20.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting ability to perform its intended function.

## 4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).
c. Packaging inspection (see 4.7).



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4.2.1 First article inspection. First article inspection shall be performed on one fan when a first article is required (see 3.2 and 6.3). This inspection shall include the examination of 4.5 and the tests of 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection. The first article may be a standard production item from the supplier's current inventory, provided the unit meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining units to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on fans selected in accordance with 4.4. This inspection shall include the examination of 4.5 and the tests of 4.6.1.

4.3 Inspection lot. All units of the same type offered to the Government at one time shall be considered a lot for purposes of inspection. The sample unit shall be one complete fan.

4.4 Sampling. Random samples shall be selected from each lot in accordance with MIL-STD-105. Any fan in the sample containing one or more defects shall be rejected; and if the number of defective units in any sample exceeds the acceptance number for that sample the lot represented by the sample shall be rejected.

* 4.4.1 Sampling for examination. A random sample of fans shall be selected from each lot offered to the Government at inspection level II. The acceptable quality level shall be 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

* 4.4.2 Sampling for tests. A random sample of fans shall be selected from each lot offered to the Government at inspection level S-4. The acceptable quality level shall be 1.5 percent defective.

4.5 Examination. Each sample shall be examined to verify compliance with this specification. The examination shall be conducted as specified in Table II.

TABLE II. Classification of defects.

| Classification | Defects |
|---|---|
| Major: | |
| 101 | Fan performance not in accordance with AMCA Standard. |
| 102 | Construction not as specified. |
| 103 | Fan wheel wrong type. |
| 104 | Bearings of wrong type or not protected with dust seals. |
| 105 | Fan housing material too thin. |
| 106 | Fan housing not sufficiently reinforced. |
| 107 | Direction of rotation or discharge not as specified. |
| 108 | Motor not of the specified type, or characteristics incorrect. |
| 109 | Drive not as specified. |
| 110 | Motor mounting incorrect. |
| 111 | No provision for belt tension adjustment. |
| 112 | Lifting attachments misplaced or not as specified. |



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TABLE II. Classification of defects. (Cont'd)

| Classification | Defects |
|---|---|
| 113 | Lubrication system not adequate. |
| 114 | Inlet, V-belt, or coupling guards not as specified. |
| 115 | Drive shaft or drive pulleys not alined correctly. |
| Minor: | |
| 201 | Mountings or fasteners loose. |
| 202 | Treatment and painting not as specified. |
| 203 | Tools and spare parts not furnished as specified. |
| 204 | Identification marking not as specified. |
| 205 | Workmanship not as specified. |

* 4.6 Tests. The first article, when furnished, and each unit selected in accordance with 4.4.2 shall be tested, and any unit failing to pass the following tests, as applicable, shall be rejected. Tests shall be conducted as outlined in the referenced documents and as herein specified.

* 4.6.1 Operational test. The fan unit shall be connected to the specified power source and operated for 10 minutes. The fan wheel, controls, motor, and accessories shall be inspected during the test for vibration, overheating, and loose parts.

4.6.2 Capacity test. The fan shall be subjected to air volume and pressure tests in accordance with AMCA Standard No. 210. The test shall be perfomed as outlined in the test code. Presentation of AMCA certification (see 3.6) may be made in lieu of performing the capacity test.

4.6.3 Electromagnetic interference suppression tests. When electromagnetic interfenence suppression is specified, the first article sample shall be tested in accordance with MIL-STD-462, to verify compliance with the pertinent provisions of MIL-STD-461.

* 4.7 Packaging inspection. The inspection of the preservation, packaging, packing, and marking of the fans shall be in accordance with the requirements of section 4 of MIL-E-16298. The inspection shall consist of the quality conformance inspection, and when specified (see 6.2.1), the preproduction pack inspection requirements of MIL-E-16298.

* 5. PACKAGING

* 5.1 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking of the fans shall be in accordance with MIL-E-16298. The level and packaging of preservation, and level of packing shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The three types of centrifugal fans covered in this specification apply to fans which are not integral parts of equipment designed as complete units by manufacturers, and are intended to be used as follows. Type 1, forward-curved fans, should be used where compactness of site is a



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factor in packaged unit applications (example: in a heating or a high efficiency air conditioning system). Type II, radial blade fans, should be used in systems handling such materials as heavy dust (example: in a woodworking shop). Type III, backward-inclined blade fans, which may be used in air conditioning applications, should be used when high efficiency and nonoverloading characteristics are required to move air containing no condensable fumes or vapors (example: in a power generating plant).

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of fan required (see 1.2).
c. When first article is required for inspection and approval (see 3.4).
d. Capacity (cfm) at the required static pressure (inches $H_2O$) (see 3.10).
e. Fan wheel material required (see 3.11.1).
f. Requirement for bearings if other than specified (see 3.11.3).
g. When bearing-cooling fan is required (see 3.11.3.1).
h. Angle of discharge if other than specified (see 3.11.4).
i. When a drain connection is to be fitted to the scroll and size required (see 3.11.4).
j. When an access door is to be installed on the fan scroll of type II fans (see 3.11.4).
k. Whether motor is to be single or multi-speed; features of bearings, enclosure, and electrical characteristics; and when an across-line partwind wye-delta or autotransformer is required (see 3.11.5).
l. When the motor shall be open, drip-proof, totally enclosed, or explosion proof (see 3.11.5).
m. When a direct drive is required, and whether resilient vibration isolators are required (see 3.11.5.1).
n. Sheave type requirements (see 3.11.5.2).
o. Type of motor starter control, and control enclosure requirements (see 3.11.5.3).
p. Requirement for air inlet guard if other than specified (see 3.12.1).
q. Electromagnetic interference suppression control is required (see 3.13).
r. When lifting devices are required (see 3.14).
s. Lubrication requirements (see 3.15).
t. When fungus resistance treatment is required (see 3.16).
u. Treatment and painting if other than specified (see 3.17).
v. Color of finish coat required (see 3.17).
w. When the preproduction pack inspection requirements are required (see 4.7).
x. Level of preservation and packaging and level of packing required (see 5.1).



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* 6.2.2 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved DID (DD Form 1664) and delivered in accordance with the CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

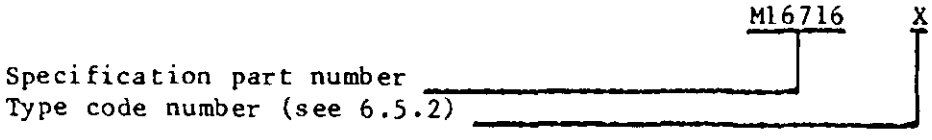
| PARAGRAPH | DATA REQUIREMENTS | APPLICABLE DD-1664 |
|---|---|---|
| 3.6, 4.6.2 | Certification of Compliance | DI-E-2121 |
| 3.18 | Publications, Commercial | DI-M-24010A |

(Copies of Data Item Descriptions required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete fan unit or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all acquisition instruments regarding arrangement for examinations, tests, and approval of the first article.

* 6.4 Definitive specification part number. The specification part number is a definitive part number which corresponds to the type of fans covered by this specification, and defines the requirements of the options presented under this specification. The specification number and the type code number are combined to form the definitive specification part number.

* 6.4.1 Cataloging data. For cataloging purposes, part numbers for the fans are assigned as follows:




* 6.4.2 Type. The type of fans (see 1.2) are identified by a single digit number (see Table III).

TABLE III. Code letter to type.

| Type | | |
|---|---|---|
| I | II | III |
| 1 | 2 | 3 |



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* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing activity:

Navy - YD

Project No. 4140-0088

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